

machine learning models. Binz and Schulz (2023a) used this dataset to fine-tune an LLM, achieving similar performance. Chen et al. (2023) built a risky choice dataset and found that GPT-3.5 makes economically rational decisions, which we replicate in task 3 of our forward modeling experiments on the `choices13k` dataset across various LLMs.

People are able to infer an agent’s beliefs, desires, and percepts from their actions (Baker et al., 2017; Lucas et al., 2014; Jara-Ettinger et al., 2020; Ho and Griffiths, 2021). These inferences are typically modeled by assuming that people employ a forward model of decision-making — typically a noisy rational model — and use Bayes’ rule to invert that model (for more details, see Section 4). The study that we focus on, by Jern et al. (2017), was intended to directly test this assumption. Similar approaches have been used to align AI systems to preferences inferred from observed user interactions (Fränken et al., 2023) or to improve human-AI interaction (e.g., Dragan et al., 2013; Sadigh et al., 2016). More generally, inverse modeling can also be viewed as inductive reasoning from behaviors to utility functions, where LLMs have been shown to be skilled at proposing hypotheses to explain observations, but not at applying these hypotheses to novel examples (Qiu et al., 2024). Lastly, recent work has extended forward and inverse models using LLMs to make inferences from utterances as well as actions (Zhi-Xuan et al., 2024; Ying et al., 2023).

3 FORWARD MODELING: PREDICTING WHICH GAMBLE PEOPLE WILL CHOOSE

Tasks. Our forward modeling experiments used the risky choice paradigm, one of the most basic and extensively studied problems in psychological decision theory (Kahneman and Tversky, 1979; Edwards, 1954). In this paradigm, participants face a choice between two gambles A and B , each with a set of outcomes that differ in their payoffs $\mathbf{R}$ and probabilities $\mathbf{Q}$. For instance, a risky choice problem might ask, “would you rather win \$5 with probability 1 or take a 0.5 probability of winning \$10?”. This can be formalized as gamble A having a single outcome with reward $\mathbf{R}_A = [5]$ and probability $\mathbf{Q}_A = [1]$, and gamble B having two potential outcomes with rewards $\mathbf{R}_B = [10, 0]$ and probabilities $\mathbf{Q}_B = [0.5, 0.5]$. Given a risky choice problem, the goal is to find a probability $P(A)$ that is consistent with how likely people would select gamble A over gamble B .

To understand how LLMs empirically capture human intent and align with actual human decisions, we designed three forward modeling tasks. First, we asked LLMs to predict the decisions that a human participant would make. Second, we asked LLMs to predict the proportion of participants that would select each option. Third, we instructed LLMs to simulate participants by making the decisions themselves. The second task corresponds directly to the original objective of finding probability $P(A)$ consistent with how likely people would select gamble A over gamble B . In the first and third tasks the LLM outputs a binary decision $\{A, B\}$, which is repeated many times to compute an aggregate probability estimate of choosing gamble A , which is compared against $P(A)$.

Human data. Human choice data came from the `choices13k` dataset (Peterson et al., 2021; Bourgin et al., 2019), a comprehensive collection of 13,006 risky choice problems. Each choice problem was answered by 15 or more participants, and participants answered each choice problem five times. For each problem, the dataset included the proportion of participant answers that selected each option. Our analyses used a subset of 9,831 problems that were not “ambiguous” (where probabilities were not shown) or lacked feedback. We used the data to evaluate the alignment of LLMs with actual human choices in each of the three paradigms.

Language models. We evaluated the following open-sourced and closed-sourced models: Llama-3-8B, Llama-3-70B, Claude 3 Opus, GPT-4-Turbo (0125-preview), and GPT-4o. We implemented experiments on the full `choices13k` dataset for zero-shot and CoT prompting across the three different tasks mentioned above for all models besides Claude. For Claude 3 Opus, we only evaluated the first task — predicting what a human participant’s choice might be — due to cost limitations.

For the experiments predicting and simulating individual human decisions (tasks 1 & 3) and models Llama-3-8B, Llama-3-70B, and GPT-4-Turbo, we conducted zero-shot experiments with n completions, where n is the number of participants that made the same decision in the `choices13k` dataset (ranging from 15 to 33). Each completion involved predicting or simulating a single human participant’s decision, with temperature set to 0.7 to maintain sample diversity. For the corresponding CoT experiments, we observed that responses were completely deterministic at temperature 0.7 on a

random subset of 1000 decisions, and thus prompted for 1 completion with temperature 0.0. For the same reason, we ran the experiments predicting the proportion of humans who would select an option (task 2) for 1 completion for both zero-shot prompting and CoT prompting with temperature 0.0. To validate our findings, we also ran ablations varying temperature, which we report in Appendix E.1.

We adapted a prompt template previously used by Binz and Schulz (2023a, see Appendix A for example prompts). For each choice problem, we first introduced the decision context of the `choices13k` dataset (i.e., the idea of choosing between gambling options) before providing each option’s probabilities and associated rewards in dollars. Finally, we asked what the participant(s) would choose (for predicting decisions) or what “you” would choose (when simulating humans). To minimize any positional biases (Wang et al., 2023), we shuffled the order of the options presented in the prompt. We also conducted an ablation where we assigned demographic profiles to the decision maker to show that our results are consistent across prompt variations (see Appendix E.2).

3.1 RESULTS

To evaluate whether LLMs are able to predict or simulate human risky choice decisions, we computed correlations and mean-squared errors (MSE) between LLMs’ responses and human decisions. Specifically, we compute the correlations across proportions where the LLM predicts that one gamble will be selected over the other. We report both Pearson and Spearman correlation, but they are extremely similar and we discuss them interchangeably. We also compared LLM responses to a classic model of rational choice: choosing the option with the highest expected value.¹ We focus here on the first of our three tasks — predicting individual choices — because results were similar across the three tasks. Results from predicting the proportion of human choices and simulating decisions are in Appendix B.

LLMs using zero-shot prompts are poor predictors of human choices. We found that LLMs with zero-shot prompting are not well-aligned with human decisions. Table 1 shows the model predictions of human behavior against real human choices, where GPT-4-Turbo performed best with a correlation of 0.60 and an MSE of 0.13. LLMs are also not well-aligned with rational decision-making. Table 2 shows that the human correlation with the maximum expected value is 0.48, while GPT-4-Turbo and 4o only achieve correlations of 0.41 and 0.28, with MSEs of 0.27 and 0.36 respectively.²

LLMs using zero-shot prompts sometimes ignore probabilities completely. To determine exactly what LLMs are doing in the zero-shot case, we fit 18 behavioral models to the responses of GPT-4-Turbo to examine its behavior. These included heuristic models (He et al., 2022), where people are thought to use mental shortcuts to make decisions, counterfactual models (He et al., 2022), which appeal to constructs like regret and disappointment, and subjective Expected Utility models, which assume that quantities involved (money and probability) are perceived or treated subjectively. This includes many of the most influential models including Prospect Theory (Kahneman and Tversky, 1979) and the model proposed by (Peterson et al., 2021), MOT.

In Appendix D, we provide the full analysis with these models. The best interpretation we found uses MOT, where the model fit a mixture of two probability weighting functions — one linear (matching humans), and one approximating a flat line. This suggests that the LLM often expected people to be rational, but completely ignores all probabilities (i.e., weighs them equally) in other cases.

LLMs using CoT assume people are more rational than they are. We found all LLMs with CoT prompting assume people act rationally, which is not aligned with actual human decisions. As shown in Tables 1 and 2, even Llama3-8B achieves a correlation with maximum expected value of 0.57, while humans only achieve a correlation of 0.48. This correlation rises as model capabilities improve: Llama3-8B obtains a correlation of 0.57 with an MSE of 0.22; Llama3-70B obtains a correlation of 0.80 with an MSE of 0.1; Claude 3 Opus obtains a correlation of 0.76 with an MSE of 0.12, while GPT-4-Turbo (which best predicted people) and GPT-4o obtain correlations of 0.93 and 0.94 with MSEs of 0.03 and 0.02. The same patterns held in the tasks asking for aggregate behavior or for LLMs to act as humans, although the aggregate behavior task had slightly reduced correlations with

¹Here, Pearson and Spearman correlation are identical as maximizing EV results in a binary response.

²In the zero-shot case, we ran a single completion for GPT-4o but multiple for GPT-4-Turbo, which likely resulted in GPT-4o being less correlated with human choices due to variance in sampling.

expected value (see Appendix B). In contrast, we also prompted GPT-4o to make the same predictions for monkeys — which were less rational, but actually closer to human choices (see Appendix E.3).

Table 1: Correlation between LLM predictions of human choices and actual human choices.

		Llama3-8B	Llama3-70B	Claude3 Opus	GPT-4-Turbo	GPT-4o
Zero-shot	Spearman correlation	0.3797	0.5300	/	0.6048	0.4756
	Pearson correlation	0.3830	0.5270	/	0.5824	0.4718
	MSE	0.1283	0.1142	/	0.1369	0.1987
CoT	Spearman correlation	0.4625	0.6156	0.5755	0.6393	0.6113
	Pearson correlation	0.4611	0.6112	0.5750	0.6326	0.6164
	MSE	0.1966	0.1633	0.1713	0.1595	0.1638

Table 2: Correlation between LLM predictions of human choices and the maximum expected value.

		Llama3-8B	Llama3-70B	Claude3 Opus	GPT-4-Turbo	GPT-4o	Humans
Zero-shot	Spearman correlation	0.1811	0.3378	/	0.4106	0.2843	0.4835
	Pearson correlation	0.1811	0.3378	/	0.4106	0.2843	0.4835
	MSE	0.3145	0.3378	/	0.2686	0.3579	0.2580
CoT	Spearman correlation	0.5665	0.7957	0.7566	0.9322	0.9444	0.4835
	Pearson correlation	0.5665	0.7957	0.7566	0.9322	0.9444	0.4835
	MSE	0.2181	0.1031	0.1228	0.0340	0.0278	0.2580

Although all the LLMs we investigated claim to be aligned with human preferences during the training, our empirical results suggest that these LLMs assume humans act more rationally than they actually do, particularly when CoT prompting is used. In these settings, LLMs correlated more highly with maximizing expected value than with human choices, demonstrating a gap between the implicit model of human decision-making assumed by LLMs and actual human behavior.

4 INVERSE MODELING: INFERRING PEOPLE’S PREFERENCES FROM CHOICES

While forward modeling allows us to predict someone’s decision given established utilities, inverse modeling is the process of inferring someone’s utilities via the decisions they make. Because forward and inverse modeling share the same theoretical framework, such inferences provide a complementary setting to evaluate the decision-making models that LLMs ascribe to humans. To formally measure these inferences, we adapt a psychology experiment from Jern et al. (2017).

Task. In the experiment, participants observed 47 decisions made by another person (the observee). In each decision, the observee is shown n groups of items $\{g_1, \dots, g_n\}$ and choose the group g_p which they prefer. Groups are populated with five distinct types of items, denoted A, B, C, D, X . For example, a particular decision could consist of two groups, $g_1 = \{X\}$ and $g_2 = \{A, B\}$, with $g_p = g_1$. Since the participant chose g_1 , they prefer obtaining X over obtaining both A and B .

Participants were asked to rank the 47 observed decisions (which all contained item X) based on how much the decision suggested the observee has a high preference for X . For instance, the decision ($g_1 = \{X\}, g_2 = \{A, B\}, g_p = g_1$) shows a higher preference for X than the naive decision ($g_1 = \{X\}, g_p = g_1$). Participants did not know the utilities of items. Thus, they were only able to determine the observee’s preferences based on features such as the number of items in each group.

The 47 decisions covered the most commonly distinguishable decision structures across items A, B, C, D, X , under the constraint that each item is included at most once per group. For simplicity, only decisions where X is part of the preferred group g_p were used. We provide a ranking of a subset of close decisions in Table 3, and a full list of 47 decisions in Appendix H.

We found that GPT-4 Turbo could not provide a valid output ranking 47 choices at once. Thus, for all LLMs, we obtained rankings by asking the LLM to perform pairwise comparisons across $\binom{47}{2}$ pairs of decisions. Pairwise outputs were limited to {stronger, weaker, tie}, and were aggregated across decisions to form a ranking. Ties were discouraged to capture small differences between decisions.